Memory Match - Flutter App

A memory matching card game built with Flutter/Dart featuring an Apple-style UI design.

Project Structure

```
memory_match/
+-- lib/
|   +-- main.dart          # Main game logic and UI
+-- test/
|   +-- widget_test.dart   # Widget tests
+-- screenshots/           # App screenshots
+-- web/                   # Web platform files
+-- android/               # Android platform files
+-- ios/                   # iOS platform files
+-- pubspec.yaml           # Dependencies
```

How to Configure and Run

Prerequisites

- Flutter SDK (3.x or later)
- Chrome browser (for web) or Android Studio / Xcode (for mobile)

Install Dependencies

```
cd memory_match
flutter pub get
```

Run on Android Emulator

```
flutter run -d android
```

Run on iOS Simulator

```
flutter run -d ios
```

Run on Chrome (Web)

```
flutter run -d chrome
```

Build for Web

```
flutter build web
```

Screenshots

Initial State

All 16 cards are face-down in a 4x4 grid. Each card has a unique color with a "?" symbol.

First Match Found

After flipping two matching cards (rockets), they stay revealed with a green border. The progress bar updates to show 1/8 pairs matched (12%).

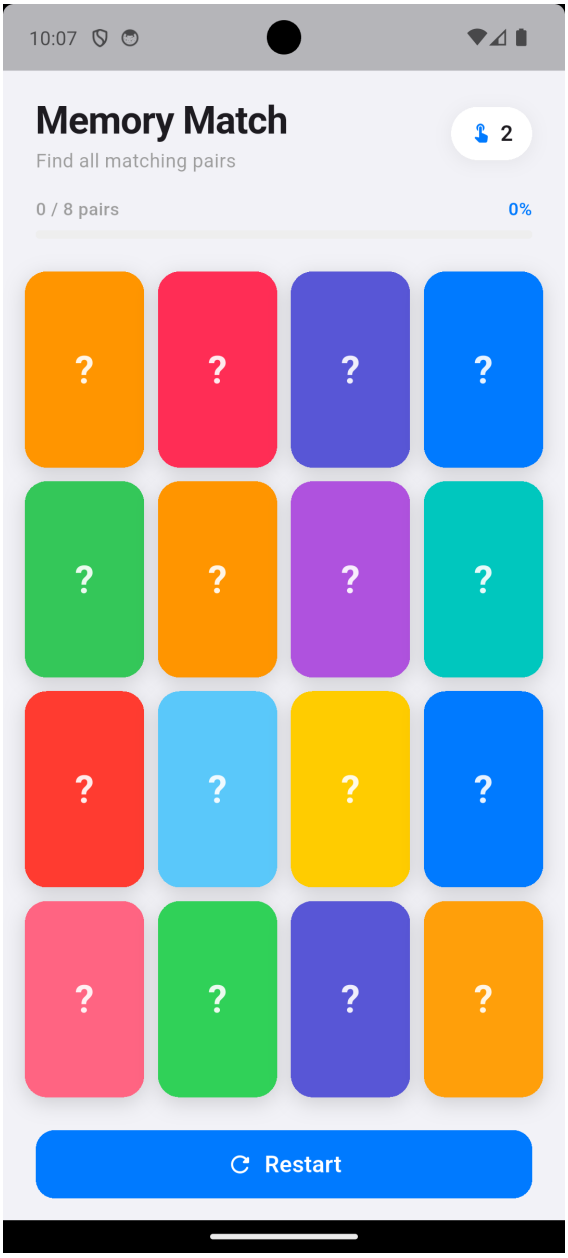

Figure 1: Initial State

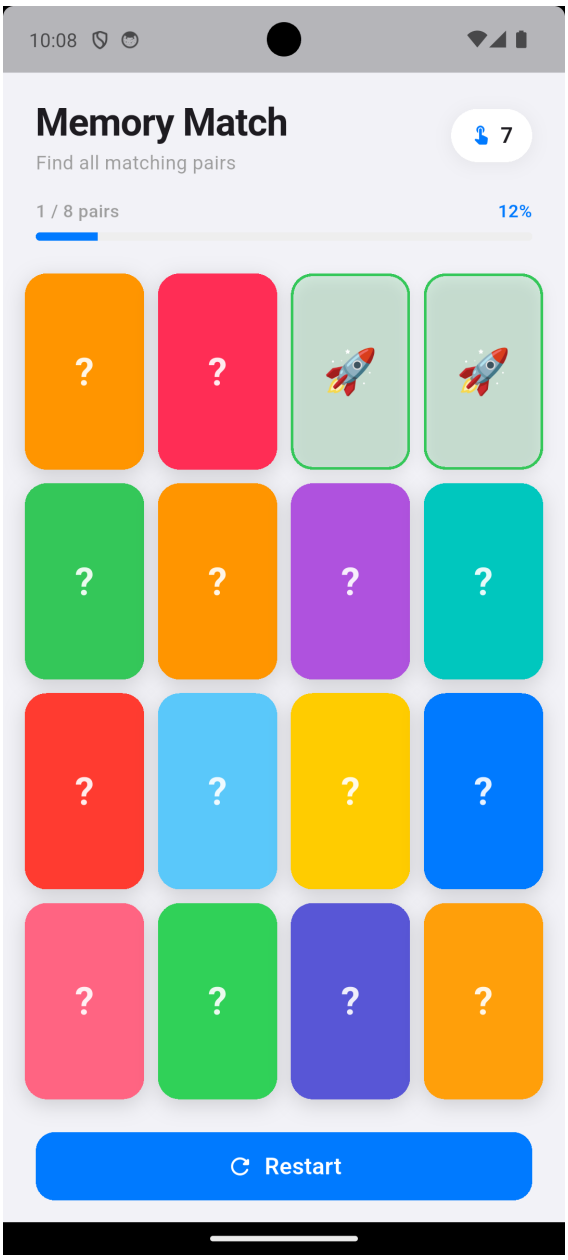

Figure 2: First Match

Multiple Matches

As the game progresses, more matched pairs are revealed. The move counter and progress bar track the player's progress.

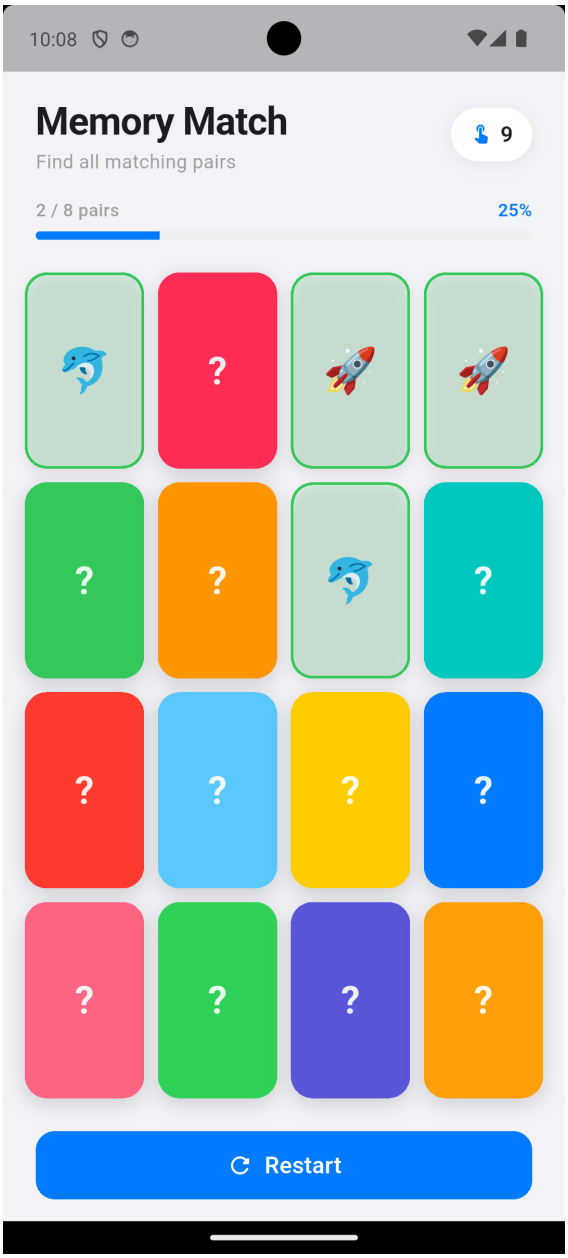

Figure 3: Multiple Matches

Gameplay Progress

The game adapts to different screen sizes with responsive layout. Here showing 3/8 pairs matched (37%).

Near Completion

With 7/8 pairs matched (87%), only two cards remain face-down. The progress bar is nearly full.

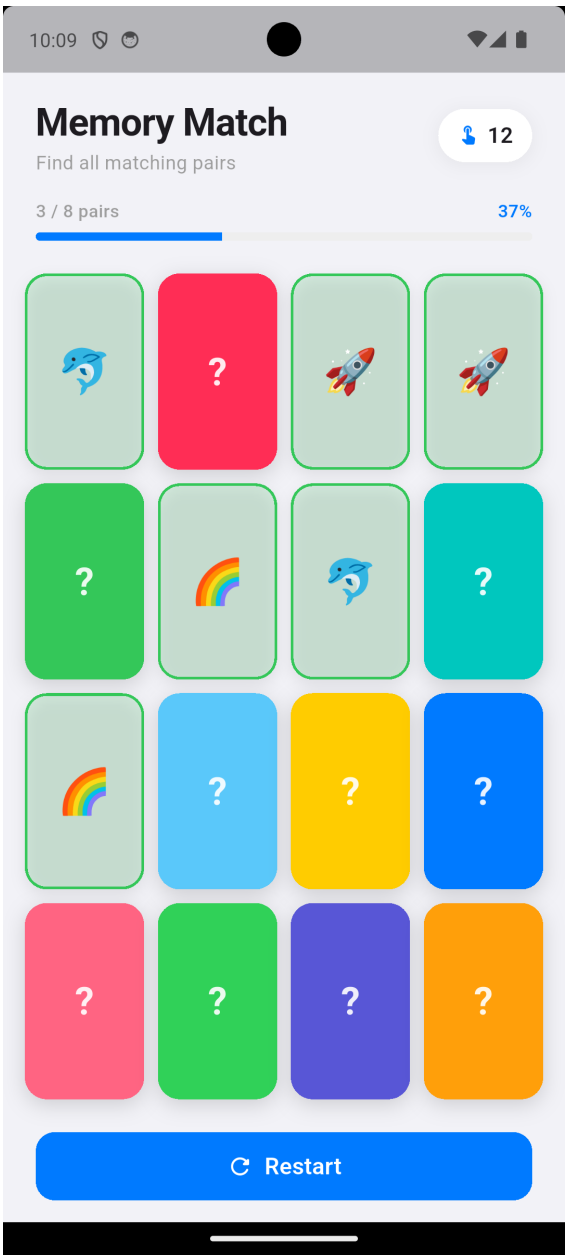

Figure 4: Gameplay Progress

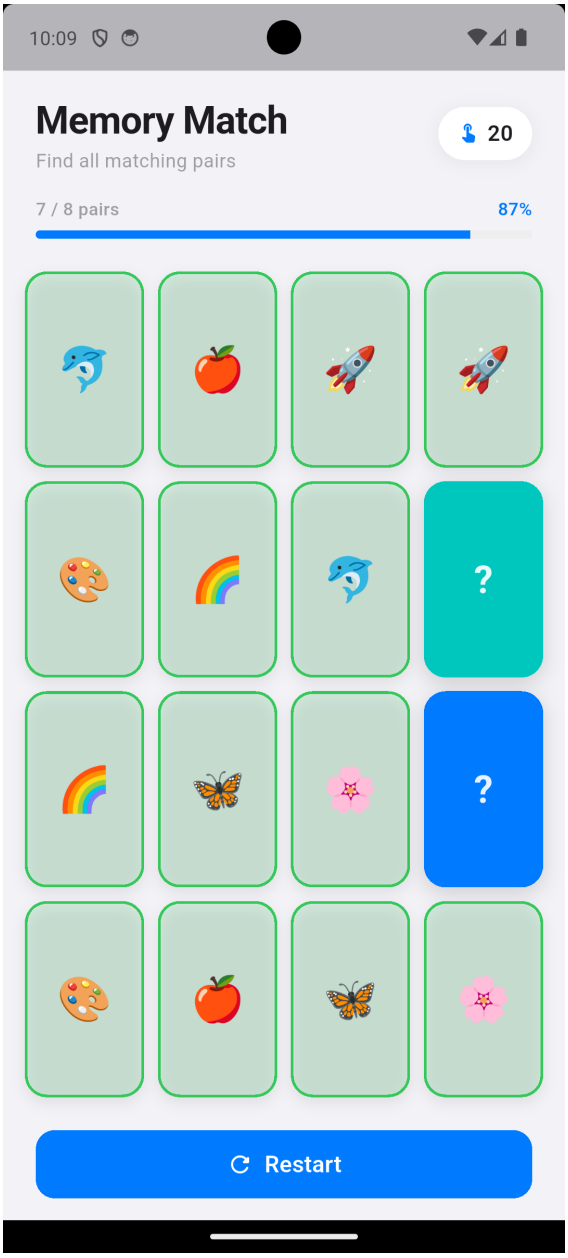

Figure 5: Near Completion

Win Dialog

When all 8 pairs are matched, a congratulations dialog appears with a blur backdrop effect, showing the total number of moves and a “Play Again” button.

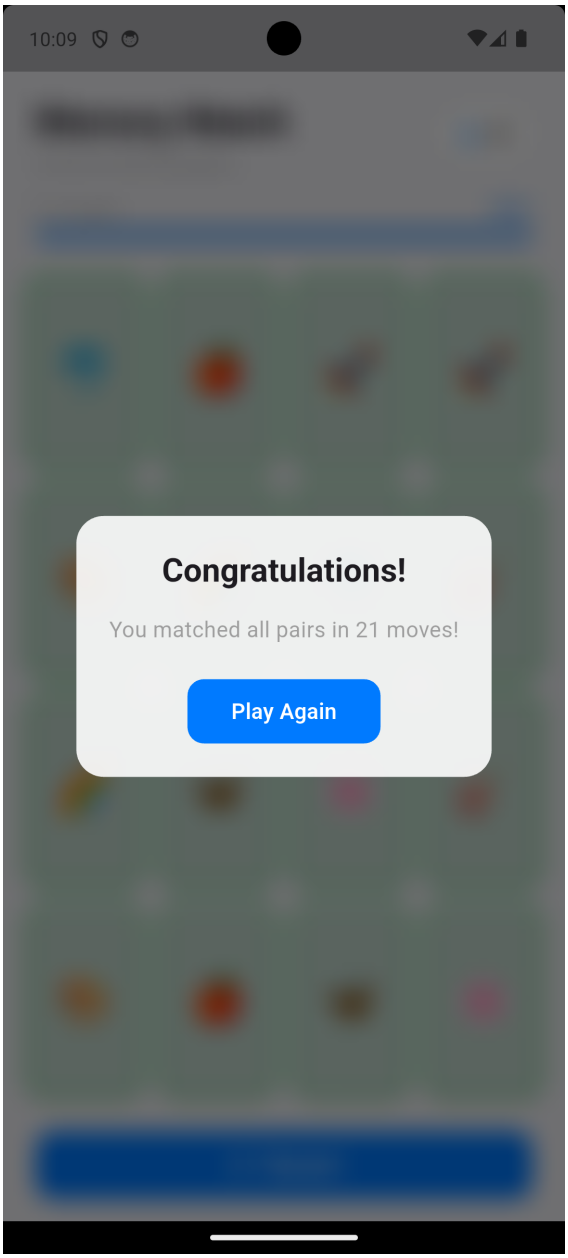

Figure 6: Win Dialog

Game Features

- 4x4 grid with 8 pairs of emoji cards
- Animated card flip transitions
- Move counter to track attempts
- Progress bar showing matched pairs percentage
- Restart button to reset the game

- Win dialog with play-again option
- Apple-style UI design (SF colors, rounded corners, clean typography)
- Responsive layout adapting to different screen sizes

Game Rules

1. All cards start face-down with colorful backgrounds
2. Tap any card to flip it over and reveal the emoji
3. Tap a second card to check for a match
4. If both cards show the same emoji, they stay revealed with a green border (matched)
5. If they don't match, both flip back face-down after 0.8 seconds
6. Find all 8 pairs to win the game
7. Try to complete the game in as few moves as possible!